Highlights

Lateral Phishing as a Violation of the Organizational Communication Graph: Cascade Dynamics and Leak-Aware Evaluation

Kamil Warpechowski, Bogdan Księżopolski

- Lateral phishing leaves a receiving-side wave on the communication graph.
- The cascade signal stays above volume detection at every stealth level.
- A time-shuffle control shows the advantage is temporal, not topological.
- The regime contrast replicates on Enron and two SNAP graphs, bounded by a timing control.
- Base-rate arithmetic makes the detectors triage instruments, not autonomous defenses.

Lateral Phishing as a Violation of the Organizational Communication Graph: Cascade Dynamics and Leak-Aware Evaluation

Kamil Warpechowski[a,b,*], Bogdan Księżopolski[a]

[a] *Kozminski University, Warsaw, Poland*
[b] *NASK National Research Institute, Warsaw, Poland*

ARTICLE INFO

Keywords:
lateral phishing
Business Email Compromise
organizational communication graph
cascade dynamics
temporal graph networks
leak-aware evaluation

Abstract

Modern email filters detect mass phishing with high accuracy (∼98%) from content and reputation, yet the residual traffic—spear phishing and Business Email Compromise (BEC)—is the most costly. Its hardest case is lateral phishing from a compromised internal account: the sender is a known contact, the address is real, and the content can be clean. We argue that such an attack is a violation of the organization's communication graph in time. The compromised account writes off-rhythm and, decisively, propagates, so the cascade leaves a receiving-side wave signature that no single message reveals.

The main result is that this signal stays above the volumetric detector under a stealthy attacker. Against an attacker who spreads the sends over time, volume-based detection—the class of methods deployed in practice—never develops a usable signal (AUC 0.48 at $K=1$; it does catch the naive blast, but that is the case the attacker can trivially avoid), while a temporal graph network reading the receiving side stays above it at every level (AUC 0.59 vs 0.48 at $K=1$); the advantage vanishes under a time-shuffle control, and the regime contrast—volume wins the naive blast, the receiving-side signal wins the stealthy regime—replicates on Enron and on two SNAP graphs (`email-Eu-core`, `CollegeMsg`), with an injected-timing control showing that once the attack paces like the background, the surviving margin belongs to the learned temporal model alone. We also mark the boundary: a single-recipient attack without propagation is invisible to every detector.

The methodological contribution is a leak-aware evaluation protocol—matched design, causal time-shuffle control, inductive organization split, label-noise audit—with explicit base-rate accounting: at a realistic prevalence each detected attack comes with hundreds of false alarms, so detectors of this class are triage instruments, not autonomous defenses. The work reframes lateral phishing as a measurable property of the communication graph: measurable, though not yet operationally solved.

1. Introduction

A mailbox is compromised on a Monday morning. Before noon, its owner's contacts in two departments and in a partner firm receive a short, plausible message with a link, each from a colleague they correspond with every week; some of them click, and an hour later their own contacts receive the same message. Nothing in any single message betrays the deception (Butavicius, Taib and Han, 2022; Sturman, Auton and Morrison, 2025). This is *lateral* phishing: a campaign sent from a genuinely compromised internal account, which Ho, Cidon, Gavish, Schweighauser, Paxson, Savage, Voelker and Wagner (2019) characterized at scale as attacks that reach many recipients inside and around the organization. FBI IC3 reports show that Business Email Compromise (BEC), of which lateral phishing is the inside-out variant, caused adjusted losses of approximately $2.9 billion in 2023 (Federal Bureau of Investigation, 2024). The sender is a known contact, the address is real, and the content is typically clean; the deception lies in *who* sends *when*, and in what happens next.

Phishing detection is often framed as an almost solved problem. Modern email filters, built on content and reputation, classify mass campaigns at ∼98% accuracy on typical benchmarks (Birthriya, Ahlawat and Jain, 2025; Tan, Chiew, Yong, Sze, Abdullah and Sebastian, 2020). However, this figure is misleading. Benchmark accuracy is dominated by the mass tail, while the targeted residual—spear phishing and BEC—is rare, drives a large share of

*Corresponding author
kamil.warpechowski@kozminski.edu.pl (K. Warpechowski); bogdan.ksiezopolski@kozminski.edu.pl (B. Księżopolski)
ORCID(s):

financial losses (Cidon, Gavish, Bleier, Korshun, Schweighauser and Tsitkin, 2019; Ho et al., 2019) while largely evading content-based methods. This residual is the core of the problem, and lateral phishing is its hardest case: a compromised account passes every check that looks at identity, reputation, or relationship, because all of them are genuine.

Our thesis is that lateral phishing is a violation of the organization's *communication graph in time*. We model the organization as a graph of who corresponds with whom, with a temporal rhythm for every account. A compromised account breaks this structure in two ways that no content filter can see. First, it writes outside its own rhythm. Second, and decisively, it propagates: the attack spreads as a cascade, and a cascade leaves a *receiving-side* signature—within a short window many accounts in one neighborhood are hit, each by a sender that was itself hit just before—which no single message reveals. This signature is what we detect, and pacing the sends weakens it far less than it weakens the sending-side burst. We are explicit about what this excludes: a single, surgical message from a compromised account to one recipient—the classic "CEO to CFO" transfer request—does not propagate and leaves no wave. It is the blind spot of every detector studied here; we measure it as such (Sec. 5.5), and it remains the domain of out-of-band process controls.

From a network-applications standpoint, corporate email is a networked service, and the organization's communication graph is an operational monitoring object of the same kind as the flow, alert, and provenance graphs used for intrusion and anomaly detection (Nayeri and Rezvani, 2026; Weng, Zhang, Zhu, Dou, Sun, Ye and Tian, 2024). The detectors studied here are accordingly designed as components of a SOC triage pipeline that scores events on that graph, not as a standalone mail filter.

We make the thesis testable through two research questions, each paired with a measurable sub-thesis.

RQ1 (signal). Can a compromised account, consistent with every static relation, be detected from cascade dynamics, and does the signal survive a stealthy attacker who spreads the sends over time? Sub-thesis T1: a temporal detector reading the receiving-side wave beats volume-based detection at low FPR and stays above it under a stealthy attacker; the advantage vanishes under a time-shuffle control and transfers in direction to real topologies.

RQ2 (limits). What are the operational limits of the signal? Sub-thesis T2: the signal needs fan-out (a single-recipient attack is invisible), degrades gracefully under label noise and scale, and at a realistic base rate the detectors serve as a triage instrument rather than an autonomous defense.

The learning models used here—a temporal graph network with node memory and gradient boosting—are standard tools; the contribution is the framing of lateral phishing as a measurable violation of the communication graph and the evidence that the receiving-side signal depends on temporal order, transfers in direction to real graphs, and stays above volume detection under stealth. Graph-based detectors are easy to inflate: real-world anchors leak rhythm, degree, or community alignment with labels, and synthetic generators can favor a model class. Following Arp, Quiring, Pendlebury, Warnecke, Pierazzi, Wressnegger, Cavallaro and Rieck (2022) and the TESSERACT protocol (Pendlebury, Pierazzi, Jordaney, Kinder and Cavallaro, 2019), every result below is produced under a leak-aware protocol with causal controls and explicit base-rate accounting, which also yields an operational definition of what it means for a detector of this class to work.

Our contributions are as follows.

1. A formalization of lateral phishing as a temporal violation of the organizational communication graph, with a cascade threat model that covers both the naive blast and the stealthy, time-spread attacker (Section 3).
2. The receiving-side cascade signal as a detection invariant: a hierarchy of detectors of increasing temporal awareness, a causal time-shuffle control, and the headline result that volume-based detection never develops a usable signal under a stealthy attacker while the receiving-side signal stays above it at every level (RQ1; T1).
3. Replication of the mechanism on three real communication graphs (Enron, `email-Eu-core`, `CollegeMsg`): the hierarchy and shuffle effect on Enron, the regime contrast on the SNAP graphs (RQ1; T1).
4. A leak-aware evaluation methodology and an operational definition of detection effectiveness for the targeted tail, with the limits stated explicitly: blind spot, label noise, scale, and base rate (RQ2; T2).

The rest of the paper is organized as follows. Related work (Sec. 2) precedes the model (Sec. 3), methodology (Sec. 4), and the case study with results (Sec. 5); the discussion (Sec. 6) follows. The complementary mode of targeted email—external impersonation of a role or authority, which requires knowledge beyond the communication graph—is out of scope here and is treated in a separate study.

2. Related Work

The review follows the chain of the thesis: lateral phishing and why content is insufficient (2.1), the graph and temporal-learning apparatus we borrow (2.2), and evaluation rigor (2.3), closing with the gaps (2.4).

2.1. Lateral phishing and BEC: why content is insufficient

Targeted attacks—spear phishing, whaling, and Business Email Compromise—differ from mass phishing in their personalization and context. A recent survey in *Computers & Security* systematizes the techniques and gaps of the area (Birthriya et al., 2025). Content-side classifiers, such as hybrid ensembles over email content and metadata (Bountakas and Xenakis, 2022), remain highly effective for the mass tail, but they inherit the content signal's blindness to clean-content, context-borne deception, and language models now make personalization cheap (Bethany, Galiopoulos, Bethany, Karkevandi, Vishwamitra and Najafirad, 2024). Ho et al. (2019) characterized lateral phishing from compromised accounts across 113 million emails and fixed an operational low-FPR threshold of <4 FP/million; their group's earlier DAS detector (Ho, Sharma, Javed, Paxson and Wagner, 2017)—a non-parametric, label-free anomaly scorer over feature vectors of clicked URLs, developed with the security team at LBNL—showed that enterprise credential spear phishing can be ranked at a practically low alert volume without any labeled training data. Cidon et al. (2019) targeted payload-free BEC in production, and BEC surveys trace its evolution and cost (Atlam and Oluwatimilehin, 2023). What the effective methods share is a departure from content: per-sender behavioral profiles (Gascon, Ullrich, Stritter and Rieck, 2018; Stringhini and Thonnard, 2015), header and stylometry models (Duman, Kalkan-Cakmakci, Egele, Robertson and Kirda, 2016), as well as analysis of relay infrastructure (Luo, Young, Ho, Schweighauser, Katz-Bassett and Cidon, 2024) all catch mail from compromised legitimate accounts whose content reads as normal. Two families of behavioral defense transfer to a compromised mail account: *per-sender deviation* from the account's own profile, as in COMPA for social-network accounts (Egele, Stringhini, Kruegel and Vigna, 2013) and in behavioral models of email senders (Stringhini and Thonnard, 2015), of which a sudden spike in volume is the simplest form; and *causal-path* analysis, as in Hopper for lateral movement in enterprise login graphs (Ho, Dhiman, Akhawe, Paxson, Savage, Voelker and Wagner, 2021), which flags a hop to a rare target without a precursor that explains it. Transferred to email headers, both read the *sending* side. We show that a stealthy attacker who paces the sends largely removes that signal, and that the receiving side of the cascade is what remains.

2.2. Graph and temporal learning for communication graphs

The cascade invariant rests on a mature apparatus of graph learning, organized here into the threads that indicate what we borrow and what is missing for the communication graph of an organization.

On the foundations of inductive learning, modern graph networks learn node representations through neighborhood aggregation: the *inductive* GraphSAGE (Hamilton, Ying and Leskovec, 2017) (generalization to nodes unseen during training) and the attention-based GAT (Veličković, Cucurull, Casanova, Romero, Liò and Bengio, 2018) establish the paradigm; our *structural* (rather than identity-based) node features are a deliberately lightweight variant of it, to preserve inductive generalization (Xu, Ruan, Korpeoglu, Kumar and Achan, 2020; Hao, Cao, Fang, Xie and Wang, 2020). Link-prediction and scalable-embedding tasks (Zhang and Chen, 2018; Zhang, Qiu, Yi and Song, 2018) provide heads for classifying $(s \to v)$ events.

The invariant requires a model of time: the memory-based TGN (Rossi, Chamberlain, Frasca, Eynard, Monti and Bronstein, 2020), the attention-based DySAT (Sankar, Wu, Gou, Zhang and Yang, 2020), EvolveGCN (Pareja, Domeniconi, Chen, Ma, Suzumura, Kanezashi, Kaler, Schardl and Leiserson, 2020), ROLAND (You, Du and Leskovec, 2022), and streaming and spatio-temporal GNNs (Yu, Xia, Liu and Wu, 2023; Wang, Chen, Xu, Wu, Shen, Xuan and Yang, 2022; Nayebi Kerdabadi, Byron, Sun and Iranitalab, 2025; Behrouz and Seltzer, 2022) capture graph evolution. Temporal graph networks are by now standard instruments for operational network security, from security-alert prediction on real networks (Nayeri and Rezvani, 2026) to real-time anomaly detection on large-scale provenance graphs (Weng et al., 2024). Our TGN-lite is a minimal realization of this paradigm by design, allowing the time-shuffle control to isolate the *mechanism* rather than the capacity of the architecture.

The receiving-side wave has close relatives in the study of diffusion on networks. Outbreak detection asks where to place sensors so that a spreading process is caught early (Leskovec, Krause, Guestrin, Faloutsos, VanBriesen and Glance, 2007); source localization infers the origin of a diffusion from which nodes were reached and when (Pinto, Thiran and Vetterli, 2012; Shah and Zaman, 2011); cascade prediction and rumor detection learn from the shape and timing of a spread whether and how far it will go (Cheng, Adamic, Dow, Kleinberg and Leskovec, 2014; Vosoughi, Roy and Aral, 2018; Bian, Xiao, Xu, Zhao, Huang and Rong, 2020). These works model a spreading process on a

known graph, as we do, but they classify the *cascade* or locate its *source*; we classify a single *event* by the state of its recipient's neighborhood, under a matched design in which every event is locally legitimate, and we ask specifically whether that signal survives an attacker who paces the spread.

The closest apparatus comes from fraud detection robust to *camouflage* and *class imbalance*: CARE-GNN (Dou, Liu, Sun, Deng, Peng and Yu, 2020), GraphConsis (Liu, Dou, Yu, Deng and Peng, 2020), PC-GNN (Liu, Ao, Qin, Chi, Feng, Yang and He, 2021a), H2-FDetector (Shi, Cao, Shang, Zhou, Zhou and Wu, 2022), GraphSMOTE (Zhao, Zhang and Wang, 2021), distribution-shift correction (Gao, Wang, He, Liu, Feng and Zhang, 2023; Tang, Li, Gao and Li, 2022), and the detection of money laundering and transactional fraud (Weber, Domeniconi, Chen, Weidele, Bellei, Robinson and Leiserson, 2019; Li, Gou, Liu, Hou, Li and Xiong, 2022a; Rayana and Akoglu, 2015). The broader graph-anomaly literature (Ding, Li, Bhanushali and Liu, 2019; Park, Kim, Han and Yu, 2020; Liu, Li, Pan, Gong, Zhou and Karypis, 2021b; Ma, Wu, Xue, Yang, Zhou, Sheng, Xiong and Akoglu, 2021; Zhang and Chen, 2021; Li, Qi, Zhao, Zheng, Cao, Dong and Yu, 2025), as well as predicting user dynamics and roles (Kumar, Zhang and Leskovec, 2019; Wu, Yuan, Lin, You, Chen, Chen and Zheng, 2020; Li, Xu, Xie, Zhou and Xuan, 2022b), provides techniques, yet operates on graphs of *transactions/reviews* and has yet to address the *propagation dynamics of a compromise* in a communication graph. Graph-based phishing detection has so far targeted mainly *web pages* and *transactions*: graph-theoretic and visual approaches for web pages (Tan et al., 2020; Lin, Liu, Divakaran, Dong, Ng, Jiang, Chen, Zhang, Lau and Dong, 2021; Abdelnabi, Krombholz and Fritz, 2020), attention over the email graph (Safran and Musleh, 2025), and propagation over URL–IP graphs (Guo, Wang, Yue, Sun and Hu, 2025; Li et al., 2022b). To our knowledge, the receiving-side dynamics of a lateral cascade have not been operationalized as a detection signal for corporate email.

2.3. Evaluation rigor: pitfalls and leak control

A central caution is that ML detectors in security are easily inflated by leaks, unfair splitting, and headline AUC in place of operational metrics (Arp et al., 2022); time-aware evaluation (TESSERACT) shows how temporal leakage distorts results (Pendlebury et al., 2019). These works motivate our leak-aware protocol and predictivity probe, which ask not how high the AUC is but whether the result transfers beyond a single corpus and whether it is an artifact of the generator. We use the same standard to define a method's effectiveness.

2.4. Gaps and positioning of the work

Several gaps emerge from the review. Detection of lateral phishing reads the sending side (volume, paths, sender profiles), which a patient attacker controls. The propagation dynamics of a compromise on the organization's communication graph have not been used as a detection signal, and evaluations rarely control for leaks or test transfer across topologies, so effectiveness tends to be ill-defined. The present work addresses these gaps: a temporal formalization of the communication graph, the cascade invariant with its causal control, replication on three real graphs, and a leak-aware methodology that defines effectiveness.

3. Model

The model formalizes the organization's communication graph in time and defines the invariant that a compromised account violates. All quantities here are variables; the numbers that instantiate them are stated in the Case Study (Sec. 5). Figure 1 gives an overview.

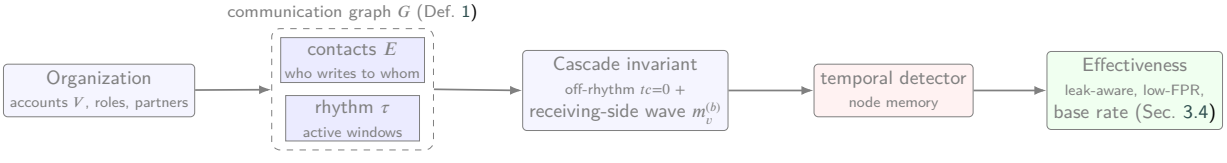


Figure 1: The model, general to detailed (left to right). An organization induces the communication graph G with a rhythm per account; a compromised account violates it by writing off-rhythm and by propagating, which leaves a receiving-side wave; a temporal detector with node memory reads that wave, and is judged by one *definition of effectiveness*.

Table 1
Notation used throughout the paper. Concrete instantiations of the free parameters (N, T, p_{inf}, H) are given in the Case Study (Sec. 5).

Symbol	Meaning
$G = (V, E, \tau)$	organizational communication graph with rhythm
V	set of accounts (people/roles), $\lvert V\rvert = N$
E	contacts: $(s, v) \in E$ iff s corresponds with v
$\tau(s)$	temporal rhythm (active windows) of sender s
T	number of time buckets partitioning the period
b	a single time bucket, $b \in \{1, \dots, T\}$
$tc(s, v, t)$	temporal consistency of (s, v) at time t
s, v	sender, victim (recipient); s_0 the compromised account
p_{inf}	per-contact infection probability of a cascade
H	maximum cascade depth (hops)
K	number of carriers of a stealthy attacker (one send per bucket)
Z	static-GNN node embedding matrix, Eq. (1)
$d_v^{(b)}$	in-degree of node v within bucket b, Eq. (3)
$m_v^{(b)}$	temporal-GNN memory of node v after bucket b

3.1. The organizational communication graph

An organization's communication is relational and rhythmic: who corresponds with whom follows the organizational chart (supervisors, coworkers, business partners), and each account is active in characteristic windows of the week. Both facts are observable from mail headers alone, without content.

Definition 1 (Organizational communication graph). *A communication graph is $G = (V, E, \tau)$, where V is the set of accounts (people/roles), $E \subseteq V \times V$ is the set of contacts ($(s, v) \in E$ iff s corresponds with v; induced by the organization's hierarchy, coworker, and partner relations), and $\tau : V \to \mathcal{T}$ assigns to every sender a* temporal rhythm*, the set of its active time windows. The temporal consistency of a message from s to v at time t is $tc(s, v, t) = \mathbf{1}[t \in \tau(s)]$.*

A message from a known contact in its rhythm, $(s, v) \in E$ and $tc = 1$, is the normal case. A known contact writing *off-rhythm*, $tc = 0$, is the canonical signature of a compromise: the relation is genuine, the timing is not. This single bit is weak on its own (legitimate mail also arrives off-hours, and an attacker with mailbox access can mimic the rhythm), which is why the invariant below reads the dynamics of the whole neighborhood rather than one message.

3.2. Threat model: the compromised account

The attacker has gained mailbox access to an internal account s_0 (the *patient zero*) and wants to deliver to victims messages classified as legitimate. From s_0 the attacker mails the account's own contacts; each recipient becomes a carrier with probability p_{inf} and propagates to its own contacts, up to depth H. The attack is therefore a *cascade* on G (Fig. 2 in Sec. 4). The term follows Ho et al. (2019), who coined *lateral phishing* for phishing sent from compromised enterprise accounts and characterized it at scale; the multi-hop cascade with a per-contact infection probability is our formalization of the compounding they observed, in which newly compromised recipients become the next senders. Every single message in it is locally plausible: the sender is a genuine contact, the address is real, and the content can be clean.

On capabilities: having mailbox access, the attacker knows the victim's rhythm and can mimic it; the attacker can also *pace* the sends—K carriers, one send per time bucket—so that no sending burst ever appears (the *stealthy* attacker, who defeats volumetric detection by construction); and the attacker can restrict the attack to a single recipient with no propagation at all (the *surgical* attack). Out of scope is an adaptive adversary with full knowledge of the graph and training poisoning, and external impersonation from outside the graph, which requires knowledge beyond G and is treated in a separate study. We treat the signal as a cost increase for the attacker; no detection is final.

3.3. The cascade invariant

Lateral phishing is a propagation process, and propagation has two sides. On the *sending* side a compromise shows up as a surge in one account's volume, or as an unusual path to a rare target; this is what practical defenses read, and it

is what a patient attacker controls. On the *receiving* side the cascade leaves a signature that no single message reveals and that the attacker cannot pace away: within the wave's window many of v's neighbors receive mail, and the sender u of the message to v was itself hit just before. This receiving-side wave is the invariant. A learned temporal model with node memory registers it by scoring an event using the state from *before* its bucket, so that it can only use the past (made precise in Sec. 4); a static model of the same topology, which sees who is connected to whom but not in what order, cannot. The invariant is testable by ablation: shuffling the time order of events destroys the order it relies on (together with within-bucket co-occurrence, Sec. 4.3), so the drop in performance under a time shuffle bounds how much of the signal is dynamics rather than topology.

3.4. Definition of detection effectiveness

Every result in the Case Study follows a standard that is detector-independent and operational; the Methodology (Sec. 4) is the protocol that operationalizes it.

A high benchmark accuracy alone does not attest to detecting lateral phishing. A method works operationally for the targeted tail only when its advantage is temporal, vanishing under a control that destroys the temporal order, and transferable, so that the qualitative hierarchy of methods—which class wins, and where it breaks down—carries over to other topologies. Effectiveness is judged by sensitivity at a low false-positive rate with explicit base-rate accounting, rather than by headline AUC. The leak-aware protocol of Sec. 4 implements this standard.

4. Methodology

The procedure turns the cascade invariant into detectors, guards against inflated results, and checks the metrics against the definition of effectiveness (Sec. 3.4). Concrete values—population, time buckets, cascade parameters, hyperparameters—are stated once, in the Case Study (Sec. 5).

4.1. Overview of the verification procedure

The procedure has four steps that mirror the model. First, instantiate the communication graph G (Def. 1) on a population, deriving the contacts E and the temporal rhythm τ. Then build detectors of increasing temporal awareness, as standard learning instruments over graph-position features, and two strong practical baselines that read the sending side. Evaluation follows under the leak-aware protocol below, so that any advantage due to leakage or capacity is destroyed by a control; the last step checks transfer to real topologies.

4.2. Leak-aware evaluation protocol

Graph detectors are easy to inflate, and the protocol below uses four safeguards against that.

A matched design makes legitimate traffic and attack edges share the same distribution of endpoints and degrees, meaning they differ only by the temporal pattern and the degree/identity confounder is removed (Arp et al., 2022). A time-shuffle control permutes the temporal order of events, which removes the advantage if it comes from dynamics and leaves it intact if it is an artifact of topology or identity (Pendlebury et al., 2019); an inductive split by organization checks that nothing is memorized about the accounts seen in training, and an audit of noisy training labels checks that the learned detector does not depend on clean labels. Operational metrics report sensitivity at a low FPR, with explicit base-rate arithmetic, rather than relying on headline AUC alone. A *predictivity probe* then asks whether what was established on one topology carries over to another (real Enron, SNAP graphs). The probe checks two properties: that the *qualitative hierarchy* of detectors—which class wins, and in which attack regime it breaks down—is reproduced on each real graph, and that the headline contrast (receiving-side signal vs. volume) keeps its direction there; with the few seeds available per real graph this is a directional check, not a significance test. We deliberately do not reduce the probe to a single rank-correlation coefficient: over a handful of detectors such a coefficient is underpowered, and the transferable claim is the regime-level contrast, not a full ranking. Together these give the working definition of effectiveness of Sec. 3.4: a method works when its advantage meets the standard defined there.

4.3. Detectors and causal controls

The design principle is uniform: features encode graph position and timing, never identity. This makes every detector inductive and applicable to accounts unseen in training.

The invariant operates on the contacts graph $G = (V, E)$ with time divided into T buckets. An attack is a cascade where a patient-zero s_0 sends malicious mail to contacts; each recipient becomes a carrier with probability p_{inf} and

propagates to depth H (Fig. 2). A single event $u \to v$ is locally indistinguishable from legitimate mail; the signal is the cascade context, since within the wave's window many of v's neighbors receive mail and u was hit just before. Four detectors of increasing temporal awareness are compared: a 1-hop tabular model (a lower bound), a hand-crafted cascade context (1-hop plus burst, freshness, and wave density), a static GNN (topology, no time), and a temporal GNN with node memory. In the stealth experiment each regime is a self-contained training task: for every stealth level K the scenario (attack plus matched benign) is generated, the split is victim-disjoint, and every detector is trained and tested *within that regime only*—no detector ever sees a mixture of regimes, so its collapse or survival is a property of the regime, not of a train–test regime shift.

The two equations below differ in one respect: whether the model may look forward in time. The static score summarizes a node by its neighbors in the whole aggregated graph, seeing who is connected to whom but not in what order. The temporal score keeps a running memory updated in event order and read from the state before the event it scores, so its memory can only use the past. This rule is enforced uniformly across the whole detector panel: every stream-derived feature—the volumetric counters, the burst/freshness/propagation windows of the hand-crafted context, and the precursor/momentum/rare-target components of the path-based baseline—is computed from events with a strictly smaller total timestamp $t(e) = (b, \text{position})$ only, where the position is the fixed (shuffled) stream order used as the tie rule within a bucket; the temporal GNN's memory is likewise updated per mini-batch of at most 64 simultaneous events in that order, each scored from the state before it, and the GNNs' static node features (in-/out-degree) are computed from the *training* split only. Two implementation pitfalls make this rule non-trivial: a naive implementation that counts over the full stream leaks future information (an event's features would see later events), and a bucket-boundary split hides the wave that builds up within a bucket, which makes the detectors asymmetric; both are avoided only by counting strictly in total event order with the tie rule above. That past-only reading is what lets the receiving-side wave become a signal, and it is what a time-shuffle destroys.

The static GNN builds embeddings from the aggregated graph with a normalized adjacency $\hat{A}$ and features X,

$$Z = \hat{A}\,\sigma\big(\hat{A}\,X\,W_1\big)\,W_2, \qquad \text{score}_{\text{stat}}(s,v) = g\big([\,Z_s,\,Z_v,\,Z_s \odot Z_v,\,x_{sv}\,]\big), \tag{1}$$

whereas the temporal GNN maintains a memory m_v and scores an event from the memory state immediately before it in event order, which preserves causality and avoids leakage,

$$\text{score}_{\text{temp}}(s,v,e) = g\big([\,m_s^{(<t(e))},\,m_v^{(<t(e))},\,m_s^{(<t(e))} \odot m_v^{(<t(e))},\,x_{sv}\,]\big), \tag{2}$$

$$m_v^{(k)} = \text{GRU}\Big(\frac{1}{d_v^{(k)}} \sum_{u:\,(u\to v)\in B_k} \sigma\big(W_m\,m_u\big),\; m_v^{(k-1)}\Big), \tag{3}$$

where B_k is the k-th mini-batch of events in total time order and $d_v^{(k)}$ the number of its messages received by v (a normalizer for the mean-pooled message).

The shuffle control is a temporal ablation. Since $\text{score}_{\text{stat}}$ is permutation-invariant in time while $\text{score}_{\text{temp}}$ is not, the drop in AUC_{temp} after time shuffling indicates how much the model relies on temporal order. The ablation is not perfectly selective: shuffling also breaks the co-occurrence of events within a bucket, so the drop is an upper bound on the contribution of order alone (consistent with the shuffled temporal model falling slightly below the static GNN, Tab. 3). We use "causal" in this ablation sense throughout, not in the sense of causal identification.

Beyond the GNN we implement two strong practical methods, both of which read the sending side.

- A volumetric baseline inspired by COMPA (Egele et al., 2013; Stringhini and Thonnard, 2015) scores an event by the sender's deviation from its own volume profile—how unusual the recipient, the time bucket, and the sender's overall activity are (three per-event features, Supplementary Sec. S6)—with the same gradient-boosting learner as the other tabular detectors,
- and a path-based baseline that transfers the causal-path idea of Hopper (Ho et al., 2021), a detector of lateral movement in login graphs, to the email graph: it looks for a causal precursor, momentum, and a rare target.

We implement their structural-temporal core rather than the full systems, which is why we label them "volumetric" and "path-based"; the exact feature set of every detector is listed in Supplementary Sec. S6, together with what the full systems add on top. Hyperparameters of all detectors are fixed uniformly, without intensive tuning, so that no detector enjoys a tuning advantage; the concrete settings are given in Supplementary Tab. S1.

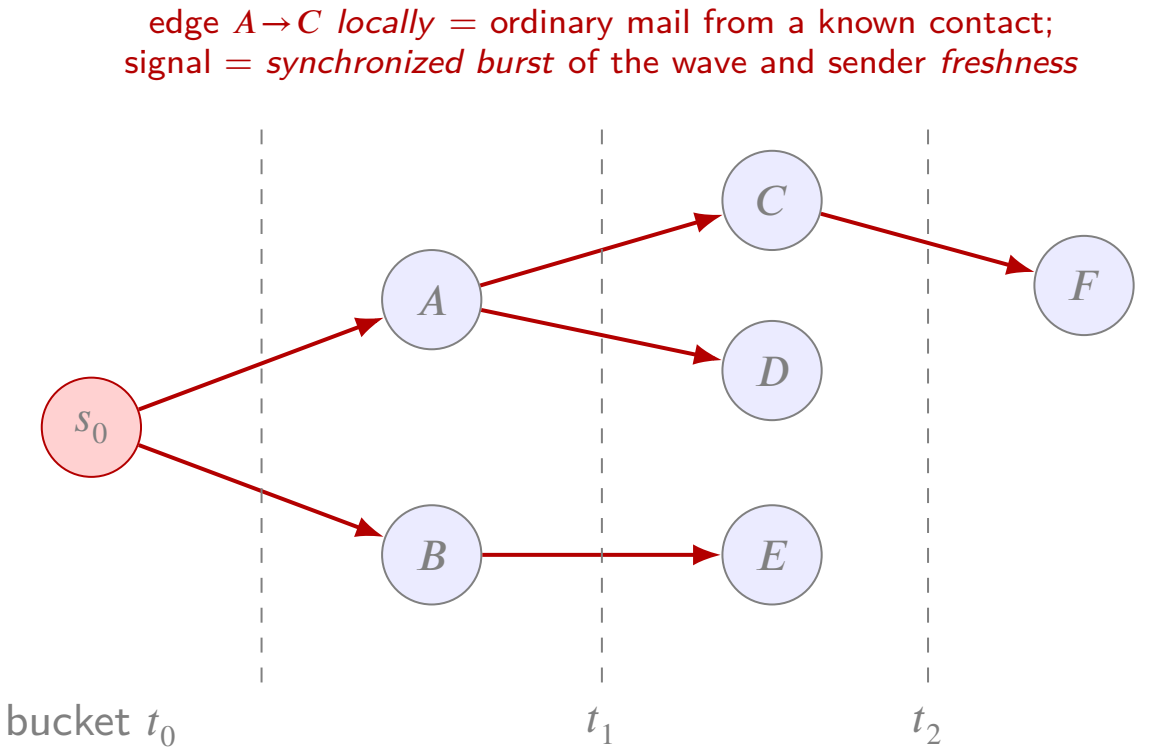


Figure 2: Lateral phishing as a temporal cascade: a carrier (s_0) infects contacts, and those in turn infect their own, so the wave propagates across the graph. Detection requires *multi-hop, temporal* propagation context.

4.4. Event model and data split

An event is a pair (sender s, victim v) with a bucket b, labeled legitimate or attack, with two-sided noise so that the classes are not perfectly separable. Legitimate events are messages between known contacts in rhythm, with a fraction arriving off-rhythm (an irreducible source of false positives). Attacks are the events of a cascade seeded at a compromised account, a fraction of which mimic the sender's rhythm. We use two split regimes, victim-wise (disjoint victim sets) and organization-wise (test organizations disjoint from training ones, an inductive validation). Headline claims are hardened over many seeds; validations on real graphs use fewer seeds, since variance is lower on a fixed structure, with a Wilcoxon test where significance matters. Seed counts for each experiment are given in the respective table or figure caption.

4.5. Metrics, base rate, and threats to validity

The headline metric is sensitivity at a fixed low FPR, the operational working point, sensitivity@FPR $= \max\{\mathrm{TPR}(\theta) : \mathrm{FPR}(\theta) \le \alpha\}$ (where $\alpha = 0.01$ for our primary analysis); AUC is reported alongside it. Real systems operate at a far stricter FPR (<4 FP/million, (Ho et al., 2019)), so the FPR reported here (an FPR of 10^{-2} against $4{\cdot}10^{-6}$) is a liberal operating point rather than a deployment threshold; at a realistic prevalence, precision follows from Bayes' theorem, which we make explicit in the Case Study (Sec. 5.8). To isolate the signal we use the temporal-order shuffle as the causal control: if the signal is truly dynamic, the control must destroy the advantage.

Threats to validity. The internal threats are label leakage, which we counter by separating features from the hidden infection state and letting the shuffle control isolate the mechanism, and dependence on the cascade parameters, which we counter by sweeping the infection probability and depth (Supplementary Tab. S2). The external threat is that the maliciousness is synthetic, so we validate the mechanism on real Enron and two SNAP graphs under controlled injections; the synthetic data lets us control the cascade at the cost of absolute effects. The construct threat is addressed by preferring sensitivity at low FPR to headline AUC, in line with (Arp et al., 2022; Pendlebury et al., 2019).

5. Case Study and Results

The *case study* defines the organizational population, the communication graph with its rhythm, and the cascade parameters that instantiate G, providing the concrete values for the symbols in Sec. 3.

5.1. Case study: the organization and what is tested on it

The case study is a fully specified organizational ecosystem (Fig. 3): 19 firms from distinct sectors (a bank, a hospital group, a software house, a law firm, a logistics company, an insurer, a public office, …), each a small office of ~ 8 accounts with one managing director, one senior manager, and five to six specialists and analysts (19/19/57/55 accounts in the four seniority levels). Everyone in a firm corresponds with everyone else in it, and the firms are tied into a ring of business partnerships (director to director, plus one specialist counterpart per partnership), which gives $N=150$ accounts and 556 contact edges (518 inside firms, 38 between partner firms). The population is synthetic, but it is generated *graph-first*: the organization graph is designed deterministically and each persona is then generated *in its node*, so that a persona's supervisor, coworkers, and external contacts are *real other personas* with real corporate addresses, and the graph closes internally. The size $N=150$ is of the order of the core user set of the Enron corpus (Klimt and Yang, 2004); the Enron contacts graph used for validation (Sec. 5.7) is larger ($\approx$1970 accounts), so the real-graph replication also tests the mechanism at a larger scale; anomaly analysis of this corpus as a communication graph has a long precedent, from scan statistics over Enron graphs (Priebe, Conroy, Marchette and Park, 2005) onward. That graph is built from the `From`/`To`/`Cc` headers of the maildir corpus as an undirected sender–recipient graph; we keep the 2000 highest-degree accounts and remove nodes left isolated by this filter, which leaves $\approx$1970 accounts. A companion study of the external-impersonation mode uses 818 senders from the same corpus: its multiplex requires a per-sender rhythm, so it keeps only senders with at least 25 messages and a non-empty active-bucket set; the difference is therefore a filter serving a different construction (topology breadth here, per-sender temporal profiles there), not an inconsistency.

We divide the week into $T=28$ buckets (7 days $\times$ 4 six-hour blocks); an account's rhythm τ is the set of its active buckets (6 per persona synthetically; on Enron derived from the `Date` headers). Realistic noise: 15% of legitimate mail arrives off-rhythm and 25% of attacks mimic the rhythm. For the cascade (Sec. 4.3) the per-contact infection probability is $p_{\text{inf}}=0.6$ and the maximum depth is $H=4$. Detector hyperparameters are fixed uniformly (Supplementary Tab. S1).

What the case study tests One kind of attack is injected into this graph, in three regimes (Fig. 3). A *compromised internal account* s_0 mails its own contacts off-rhythm; each recipient becomes a carrier with probability p_{inf}, so the attack spreads as a cascade that crosses into the partner firm through the business-partnership edge. The sender is a genuine contact consistent with every static relation, so only the timing and the receiving-side wave give it away, and this is what T1 must catch. Beyond the default cascade we probe three *extreme* regimes, each chosen to stress a specific claim rather than to represent typical traffic. *(i) Naive blast*: one carrier infects all contacts at once, the maximally bursty case and the easy end of the scale—with event-causal counters the volumetric detector *does* catch it (AUC 0.83 on the SNAP blast regimes, Tab. 6), which is exactly what makes the stealthy regime the informative one. *(ii) Stealthy spread* (K carriers, one send per interval, down to $K=1$): the burst is smeared out, so this is the case that *breaks* volume detection and isolates the receiving-side signal. *(iii) Single-recipient surgical attack* ($p_{\text{inf}}=0$, fan-out 1): no propagation at all, the *blind spot* that no cascade detector can see, included to mark the lower boundary of the threat model (T2). The stealthy-attacker and baseline contrasts are averaged over 20 random seeds on synthetic data and the detector hierarchy over 5; real-graph validations use 8 (Enron) and 3 (SNAP) seeds. Seed counts are stated in every caption.

5.2. Results overview

Table 2 maps the two research questions of Sec. 1 to the evidence; the subsections follow that order. Results are reported on a victim or organization split, as AUC and, where the operating point matters, sensitivity@FPR1%; the time-shuffle control is applied to the detector hierarchy on the case-study graph and on Enron. Seed counts are stated in each table or figure caption.

5.3. The cascade detectors

A compromised account passes every static relation, so static graph position alone carries little operational signal (the static GNN reaches AUC 0.76 but only 0.039 sensitivity at FPR 1%). The signal must therefore lie in the dynamics. Table 3 tests T1 on the default cascade of Fig. 3 and shows a clear hierarchy among detectors (bar chart in Supplementary Fig. S1): local 1-hop features are essentially useless (AUC 0.70, sensitivity@FPR1% ≈ 0), while a

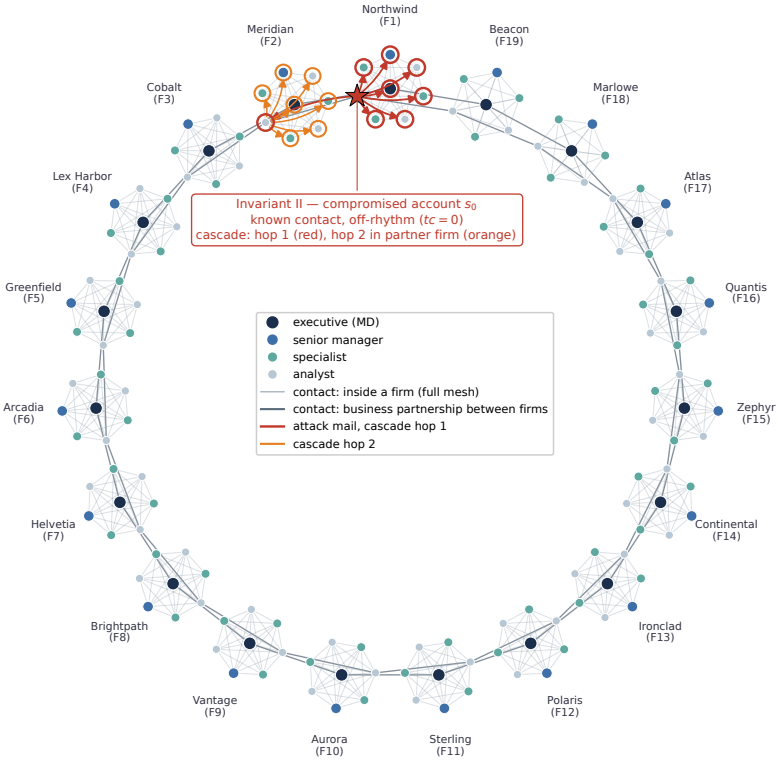


Figure 3: The case-study organization, drawn from the population data of the reproducibility package: 19 firms on a ring of business partnerships, 150 accounts colored by seniority, and the real contacts layer (556 edges). Overlaid is the attack the detectors must catch: a compromised account s_0 (star) whose off-rhythm mail cascades through its own firm (hop 1, red) and into the partner firm through the business-partnership edge (hop 2, orange). Every message in the cascade comes from a genuine contact; only the timing and the receiving-side wave reveal it.

	Sub-thesis	**Main evidence**	**Verdict**
RQ1	the receiving-side cascade signal detects a compromise and stays above volume under a stealthy attacker (T1)	Tabs. 3–5(a); real graphs: Tab. 6; Secs. 5.3–5.5, 5.7	stays above volume at every stealth level; vanishes under time shuffle; replicates in direction on three real graphs
RQ2	the signal needs fan-out ≥ 2, degrades gracefully, and is a triage instrument at a real base rate (T2)	Tab. 5(b); Supplementary Tabs. S3–S4; Tab. 8; Secs. 5.5, 5.6, 5.8	fan-out ≥ 2 needed; graceful under noise and scale; triage, not autonomous defense

Table 2
Map from research questions to evidence. Each verdict is established in the section listed.

Detector	AUC	sensitivity@FPR1%
1-hop (tabular)	0.70	0.01
Static GNN	0.76	0.04
Hand-crafted context (tab.)	0.81	0.02
Library TGN (PyG, 3 seeds)	0.79	0.11
Temporal GNN	0.87	0.10
Temporal (time shuffled)	0.73	0.02

Table 3
Cascade detectors on the case-study graph (victim split, 5 seeds; library TGN row: 3 seeds): only the temporal models with node memory reach operational sensitivity (0.099 and 0.112 at FPR 1%), the volumetric baseline ranks the naive cascade well (0.83) but converts it to a worse operating point than the temporal GNN (0.23 vs 0.10 at a much noisier threshold), and shuffling the time order removes the temporal advantage ($\rightarrow$0.73), so the signal depends on temporal order.

hand-crafted multi-hop context reaches 0.81. A static GNN captures the topology but misses the timing (0.76), and the volumetric baseline—evaluated in the same pass—ranks the naive cascade at 0.83 but converts it into a worse operating point than the temporal model (0.23 vs 0.10 sensitivity at FPR 1%), leaving the temporal GNN in the lead (AUC 0.87, sensitivity@FPR1% 0.099). This lead is specific to the default cascade: under the stealthy regime of the second harness the temporal GNN and the hand-crafted context are statistically indistinguishable (Sec. 5.8). This advantage is temporal in the ablation sense. A simple time permutation drops the temporal GNN back to the static level (0.87 $\rightarrow$ 0.73, sensitivity 0.099 $\rightarrow$ 0.022). Architecture choice matters less than the paradigm; TGN-lite and TGAT-lite outperform the hand-crafted context (0.87 and 0.88 vs 0.81 AUC; Supplementary Sec. S1), and the library TGN (Rossi et al., 2020) reaches 0.79 AUC with 0.112 sensitivity@FPR1%—below our TGN-lite on AUC but close to it on the operational metric (the library-TGN row of Tab. 3, 3 seeds).

5.4. The stealthy attacker

Table 4 is the headline test of T1: the stealthy attacker breaks volume, but not dynamics. Against an attacker who evades detection by spreading its sends over time (K carriers, one per interval), the volume detector never develops a usable signal (AUC 0.50–0.63 across all levels—its event-causal counters now see the burst that builds up within a bucket, but one send per bucket leaves every causal window empty), while the temporal GNN stays above it at every level (AUC 0.59, sensitivity@FPR1% 0.026 vs 0.000 for volume at K=1) because it picks up the receiving side of the cascade. Each detector is trained and tested within a single K regime on a victim-disjoint split (Sec. 4.3), so the volumetric weakness is not a train–test regime shift. We also report a rhythm-only control: in this harness the off-rhythm share is nearly identical for attacks (75–79%) and for legitimate traffic (78–79%), because the benign buckets are drawn uniformly and the attack buckets consecutively, so the rhythm bit alone stays at chance (0.50–0.54) and none of the signal can be attributed to it; the 15%/25% rhythm parameters quoted in Sec. 5.1 belong to the companion study's persona generator, not to this harness. At K=1 the temporal signal is weak in absolute terms, and the hand-crafted context is the strongest single detector (0.73, sensitivity 0.11); the claim is the relative one—the receiving-side signal does not collapse—which Table 5 then qualifies. The temporal sensitivity column is not monotone in K (0.026 $\rightarrow$ 0.051 $\rightarrow$ 0.042 $\rightarrow$ 0.044 from K=1 to K=5) while the corresponding AUC rises monotonically from K=2 on (0.74 $\rightarrow$ 0.74 $\rightarrow$ 0.76); we attribute the non-monotonicity to seed-level variability of the low-FPR operating point, which depends on a tail quantile of the score distribution, rather than to a change in mechanism.

Figure 4 plots the same breakdown across K: both detectors weaken as the attack gets stealthier, and the receiving-side signal keeps its margin over the volumetric one at every level.

5.5. Baselines and the boundary of the threat model

Table 5 places this result against the two strong practical baselines and marks the boundary of the threat model. This table comes from the second experiment harness (Supplementary Sec. S6), so its K=1 values differ slightly from Tab. 4. In the stealthy regime (block a) the path-based detector and the hand-crafted context lead under extreme stealth (K=1: 0.74 and 0.77, ahead of the temporal GNN 0.63) but degrade with reach (path 0.74 $\rightarrow$ 0.65), while the temporal GNN has the largest end-to-end gain (0.63 $\rightarrow$ 0.85) and leads from K=3 on. The two harnesses differ in the generator and the population, not in the feature definitions: Tab. 4 uses the 150-account case-study graph with consecutive-bucket stealth and uniform benign buckets, while Tab. 5 uses a 600-account synthetic organization with

	AUC			sensitivity@FPR1%		
Level	vol.	manual	temp.	vol.	manual	temp.
K=1 (extremely stealthy)	0.48	0.73	0.59	0.00	0.11	0.03
K=2	0.55	0.70	0.74	0.01	0.02	0.05
K=3	0.58	0.65	0.74	0.01	0.02	0.04
K=5	0.60	0.67	0.76	0.03	0.02	0.04
blast	0.63	0.74	0.85	0.03	0.02	0.07
rhythm-only bit (all levels)		0.50–0.54			0.000	

Table 4
Stealthy attacker on the case-study graph (K carriers spread over time; 20 seeds): the volumetric detector is near chance at every level (0.48–0.63), the temporal GNN stays above it at every level, and the hand-crafted context is the strongest detector at K=1 (0.73, sensitivity 0.11)—the headline result behind T1 is the receiving-side signal, read equally well by the learned model and by the hand-crafted features, not the superiority of either. The rhythm-only control stays at chance at every level, so the signal is the receiving-side wave, not the rhythm bit.

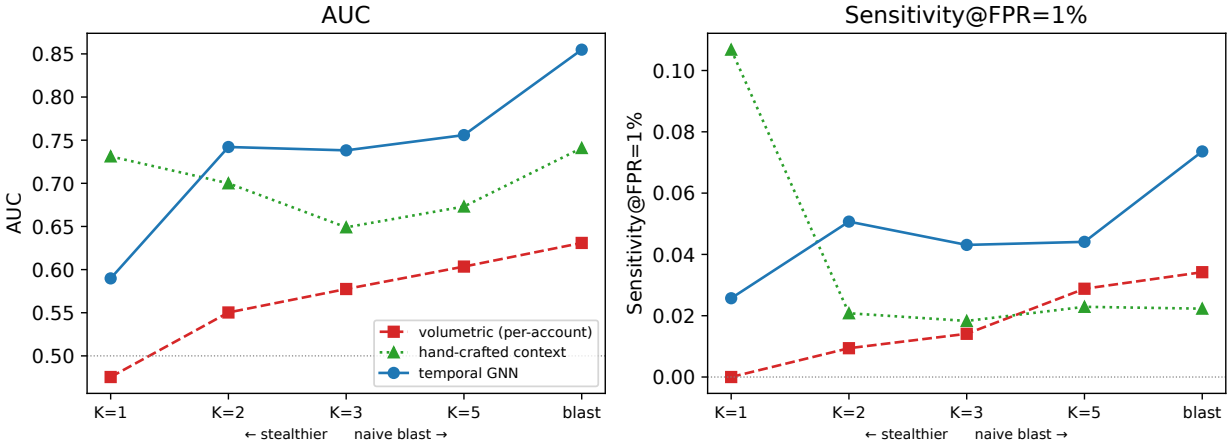


Figure 4: As the attack gets stealthier (smaller K) both detectors weaken, but the temporal GNN keeps its margin over the volumetric one thanks to the cascade's receiving-side signal.

an 8-bucket send spacing, 30% rhythm mimicry, and 15% off-rhythm benign traffic, plus a benign matched to the attack (Sec. 4.3). Two remarks on reading this table. First, the path-based baseline keeps its score although the look-ahead is removed: the corrected momentum counts *past* targets—a different signal, not a subset of the uncorrected one—and the harness uses its own generator, whose benign protocol (15% off-rhythm, spread 8, mimicry 0.3) differs from the first harness, so part of the difference against the stealth table is a harness difference, not only the causal fix. Second, the K=1 row has no single winner among path-based and hand-crafted detectors (0.74 vs 0.77); neither should be read as the regime's best detector without per-seed intervals. The ablation column "path (no precursor)" answers the question of how much of the path baseline's stealth performance is carried by its precursor feature, which largely coincides with the freshness bit of the hand-crafted context: zeroing it out drops the path baseline by 0.06–0.12 AUC at every stealth level ($0.74 \to 0.62$ at K=1, $0.82 \to 0.79$ at blast) but leaves the surgical regime untouched ($\approx$0.50, where the feature is already uninformative), so the precursor is the path baseline's stealth signal, not its surgical one.

In the surgical regime (block b, no propagation at all, p_{inf}=0) a single-recipient attack is undetectable by every detector (AUC 0.48–0.53): the signal requires fan-out $\geq$ 2, and this is the measured lower boundary of what the detectors can see (T2). Block (b) also shows that propagation itself is not required: with p_{inf}=0 and fan-out $\geq$ 2 the attack is caught by the volume detector (0.68–0.83) and by the temporal GNN (0.75–0.94), because several recipients in one neighborhood hit at once already form a synchronized sending burst and a receiving-side wave. What the signal needs is fan-out, not a cascade; the cascade matters because it is what lets a stealthy attacker reach many victims *without* such a burst. In this regime the hand-crafted context is structurally near-blind ($\approx$ 0.55 at every fan-out): its

Regime / level	vol.	path	path (no precursor)	manual	temp.
(a) Stealthy cascade: K carriers, one send per bucket, p_{inf}=0.6 (reach in parentheses)					
K=1 (11%)	0.50	0.74	0.62	0.77	0.63
K=2 (28%)	0.59	0.72	0.64	0.71	0.69
K=3 (46%)	0.59	0.66	0.63	0.66	0.71
K=5 (63%)	0.64	0.65	0.64	0.69	0.78
blast (75%)	0.70	0.82	0.79	0.76	0.85
(b) Surgical attack without propagation: p_{inf}=0, by fan-out					
fan-out 1	0.50	0.48	0.48	0.53	0.51
fan-out 2	0.68	0.50	0.50	0.55	0.75
fan-out 3	0.76	0.52	0.52	0.54	0.85
fan-out 5	0.83	0.50	0.50	0.52	0.94

Table 5
Baselines and the boundary of the threat model (AUC, 20 seeds; path-based baseline in the spirit of Hopper (Ho et al., 2021); "path (no precursor)" zeroes the precursor feature of the path baseline). (a) Under a stealthy cascade the path-based detector and the hand-crafted context lead only at K=1, the path-based detector degrades with reach, and the temporal GNN has the largest end-to-end gain, leading from K=3 on; removing the precursor costs the path baseline 0.06–0.12 AUC at every level. (b) With no propagation, a single-recipient attack is invisible to every detector (0.48–0.53); the signal needs fan-out ≥ 2. The two blocks are not comparable at "1": in (a) K=1 still propagates, in (b) fan-out 1 is a surgical attack with no cascade at all.

burst feature counts the *victim's* contacts hit in strictly past buckets, but a surgical carrier hits each of its own contacts once at staggered buckets, so a victim's neighborhood is not revisited within the window; the volume detector sees the carrier's own multi-send pattern and the temporal memory accumulates the staggered sends.

Legal broadcast waves as hard negatives. A cascade detector that fires on *any* wave would be useless in practice, because real organizations contain legal broadcast waves (newsletters, HR announcements, incident communications). We inject two classes of them (20 seeds; `exp_legal_cascade.csv`). *Easy waves*: contact-preserving, in-rhythm, fan-out 2, 30% of the attack volume. *Hard waves*: deliberately off-rhythm, fan-out 6, three hops through the contact graph including the inter-firm partnership edges—structurally as wave-like as the attack, and differing only in intent. Both cost almost nothing in the default cascade: volume moves 0.834 → 0.830 (easy) → 0.817 (hard), hand-crafted 0.805 → 0.801 → 0.792, temporal 0.868 → 0.870 → 0.879—the learned model is *more* accurate with the extra negatives to learn from. Under the stealthy attack the same holds with the hard waves present: temporal 0.746 vs volume 0.534 and hand-crafted 0.653. The legal waves are sender-initiated and paced like the background, so they look exactly like what the receiving-side features expect of legitimate traffic; the contrast with the stealthy *attack*—which hits the same contact pattern but paces its sends to hide the burst—comes from timing, not topology. The measured boundary of the threat model is therefore two-sided: the detectors do not need a malicious cascade to fire, but they also do not fire on a merely synchronized legal one, even off-rhythm and multi-hop.

5.6. Stability and inductive generalization

Two checks bound the reading of T1. The cascade model sweep (Supplementary Tab. S2) shows the hierarchy holding across the whole range $p_{\text{inf}} \in [0.05, 0.9]$ and depth; the receiving-side signal does not vanish at a realistically low p_{inf} (temporal GNN 0.84–0.97 across the sweep, with the volumetric baseline a solid mid-table 0.82–0.88). The temporal AUC is highest at the lowest p_{inf} (0.97 at 0.05) because the class composition changes along the sweep: the benign volume is fixed, while the number of attack events grows with p_{inf} (more infections mean more carriers and more sends), so at a low p_{inf} the attack class is small and concentrated in patient-zero bursts that are nearly separable, and it grows more benign-like as propagation strengthens. Under inductive validation across disjoint organizations (Supplementary Tab. S3 and Fig. S2), the temporal GNN does *not* lead on AUC—it is behind the hand-crafted context at every scale (0.72 vs 0.79 at 19 organizations, 0.70/0.73 vs 0.77/0.82 at 30 and 200) and, with the volumetric baseline included in the same pass, volume leads the inductive setting outright: 0.83–0.85 AUC and sensitivity@FPR1% of 0.24–0.33, against 0.09 for the temporal GNN at 19 organizations. The honest reading is that T1's inductive boundary is here: the receiving-side wave wins in-distribution and under the stealthy attacker, but its per-sender deviation features

make the volumetric detector the best *transfer* detector to organizations never seen in training, while the temporal GNN still holds a 1.8–2.3× operational edge over the hand-crafted context at every scale.
Cascade dynamics thus overcome the blindness of static relations to a compromise and, unlike the volume detector, do not collapse under a stealthy attacker; the advantage depends on temporal order, but it does not transfer across organizations as well as the volumetric signal does. The next question is whether the mechanism is a property of the generator or of communication graphs.

5.7. Validation on real communication graphs

Real graphs (Enron, two SNAP graphs) are used to validate the mechanism where the structure and rhythms are real. The qualitative hierarchy of methods established on one topology must carry over to another, in line with the definition of effectiveness (Sec. 3.4). The attack labels are controlled injections under a matched design, so legitimate traffic and attack are edges with the same distribution of endpoints and degrees, distinguished only by their temporal pattern.

Table 6 reports both replications, with the volumetric detector evaluated on all three real graphs. On the real Enron contacts graph (≈1970 accounts, 8 seeds, leak-free; block a) the regime contrast is the same as on the SNAP graphs: under the naive blast the volumetric detector wins (0.87, sensitivity 0.22)—its event-causal counters see the synchronized burst building up within a bucket—with the receiving-side hierarchy behind it (hand-crafted context 0.77, temporal GNN 0.81; bar chart in Supplementary Fig. S4) and the time shuffle leveling the temporal detector toward the hand-crafted result (0.75). Under the stealthy attacker the picture flips: the hand-crafted context reaches 0.89 (sensitivity 0.27) and the temporal GNN 0.84 (0.16), while volume stays at 0.67 and the 1-hop baseline at 0.59; the shuffle control costs the temporal detector 0.10 AUC, so the advantage depends on temporal order here as well. On `email-Eu-core` (Paranjape, Benson and Leskovec, 2017) and `CollegeMsg` (Panzarasa, Opsahl and Carley, 2009) ((Leskovec and Krevl, 2014); block b, bar chart in Supplementary Fig. S3) the same contrast replicates: under the stealthy attacker the hand-crafted context reaches 0.84/0.82 and the temporal GNN 0.79/0.83, while the volumetric detector stays at 0.62–0.63 (the better receiving-side detector is +0.16/+0.20 AUC ahead); under the naive blast the volumetric detector wins on both graphs (0.83) while the receiving-side detectors drop to 0.61–0.65. The blast inversion has a structural explanation that the two methods share: in the real-graph blast all sends of a carrier share one bucket, so the past windows of hop-1 events are empty and the receiving-side wave leaves no readable trace there, whereas on the dense case-study cliques the same burst stays visible through the contacts feature; under the stealthy cascade the staggered buckets feed the memory. The predictivity probe is therefore positive in the regime-level sense of Sec. 4.2: on each of the three real graphs the stealthy regime puts the receiving-side detectors ahead of volume exactly where the attacker hides, and the blast is volume's best case everywhere. With 3–8 seeds per graph a per-graph Wilcoxon test cannot reach conventional significance (the minimum two-sided p is 0.25 for n=3), so we report the real-graph replications as directional confirmations, not as significance tests. One caveat we quantify in Sec. 5.8: part of the receiving-side margin in the stealthy regime rides on the *injected-attack timing signature*—the deterministic send gaps of the injected events differ from the background's timing—so we repeat the stealthy regime with attack times drawn from the background's empirical gap distribution; the receiving-side signal survives in the learned temporal model, but not in the hand-crafted features. The practical takeaway of the real-graph block is therefore not "volume never works" but "volume fails exactly in the stealthy regime, which is the threat model this paper is about—and even there, only the learned memory reads the wave once the injection signature is removed."

5.8. Rigor: what it means for a detector to work

The results follow the definition of effectiveness (Sec. 3.4) and the operational limits are marked, so the headline numbers read as a lower bound rather than a deployment claim.

Variance and significance. The stealthy-attacker, baseline, and label-noise contrasts are averaged over 20 seeds on synthetic data; the detector hierarchy (Tab. 3) uses 5, the cascade sweep (Supplementary Tab. S2) 8, the inductive validation (Supplementary Tab. S3) 5 or 3, the sensitivity–FPR curve (Supplementary Fig. S5) 10, and the real-graph replications (Tab. 6) 8 (Enron) and 3 (SNAP); aggregate values for every table are available in the reproducibility package (`code/results/*.csv`), and per-seed values are dumped as `*_perseed.csv` files by every headline script (bootstrap 95% CIs in `code/results/ci/`). Strengthening the main contrast on 20 seeds at the stealthy point of the second harness (K=2, spread 8, mimicry 0.3; Sec. 5.5) with a paired Wilcoxon test and Cliff's δ, the receiving-side signal beats the volume detector significantly and with a large effect (temporal GNN +0.115 AUC over COMPA, $p < 0.001$, δ=0.98), the hand-crafted context beats it as well (+0.095, $p < 0.001$, δ=0.91), while the temporal GNN and the hand-crafted

Graph (regime)	Detector	AUC	sensitivity@FPR1%
(a) Enron contacts graph, ≈1970 accounts, matched design, 8 seeds			
naive blast			
	1-hop (tabular)	0.66	0.02
	Hand-crafted context (tab.)	0.77	0.04
	Volumetric (COMPA)	0.87	0.22
	Temporal GNN	0.81	0.07
	Temporal (time shuffled)	0.75	0.05
stealthy attacker			
	1-hop (tabular)	0.59	0.03
	Hand-crafted context (tab.)	0.89	0.27
	Volumetric (COMPA)	0.67	0.00
	Temporal GNN	0.84	0.16
	Temporal (time shuffled)	0.75	0.15
(b) SNAP graphs, naive blast vs stealthy attacker, 3 seeds			
	volumetric	0.83	0.10
`email-Eu-core` (blast)	hand-crafted context	0.62	0.02
	temporal GNN	0.61	0.02
	volumetric	0.62	0.00
`email-Eu-core` (stealthy)	hand-crafted context	0.84	0.10
	temporal GNN	0.79	0.11
	volumetric	0.83	0.12
`CollegeMsg` (blast)	hand-crafted context	0.62	0.02
	temporal GNN	0.65	0.03
	volumetric	0.63	0.01
`CollegeMsg` (stealthy)	hand-crafted context	0.82	0.08
	temporal GNN	0.83	0.09

Table 6
Replication on three real communication graphs, with the volumetric detector on all three (AUC / sensitivity@FPR1%). (a) On Enron, under the naive blast the volumetric detector wins (0.87) and the case-study hierarchy follows behind it; under the stealthy attacker the receiving-side signal wins (0.89 hand-crafted context, 0.84 temporal GNN vs 0.67 for volume), with the time shuffle costing the temporal detector 0.10 AUC. (b) The same regime contrast replicates on both SNAP graphs: volume wins the blast (0.83), the receiving-side detectors win the stealthy regime (0.82–0.84 vs 0.62–0.63). The stealthy-regime numbers embed the injected-attack timing signature quantified in Sec. 5.8 (injection control).

context are statistically indistinguishable at this point (+0.021, p=0.19, δ=0.26). We also repeated the contrast under the benign protocol of Tab. 5 (15% off-rhythm traffic): the margins are stable there (+0.109 over COMPA, $p < 0.001$; temporal vs manual +0.027, p=0.04), so under off-rhythm benign traffic the learned model gains a small but nominally significant edge over the hand-crafted features—the honest reading is that the two receivership detectors are near-equivalent, with the balance tipped slightly toward the learned model when the background is noisier. Under a stealthy attack, then, it is the receiving-side signal—read equally well by the learned temporal model and by the hand-crafted features—that surpasses volume detection. For the whole curve, instead of a single threshold we report the sensitivity–FPR curve (Supplementary Fig. S5): at a realistically low FPR (0.1%) the sensitivities fall to 0.005–0.040 in the blast regime and 0.001–0.023 in the stealthy regime.

The volumetric weakness is structural, not a tuning artifact. The volumetric baseline has no window or threshold to tune: its three features are counts over the observed stream (recipient novelty, time novelty, sender activity; Supplementary Sec. S6) fed to the same learner as every other tabular detector. Its counters are event-causal—an event's features may use earlier events of the same bucket, so a burst that builds up within a bucket *is* visible to it (this is why it wins the naive blast on the SNAP graphs, Tab. 6). Its weakness under the stealthy attacker is therefore a property of the attack model—one send per bucket per carrier leaves every causal window without a surge to see—rather than of a baseline configuration.

The injected-attack timing signature, and what survives its removal. The stealthy regimes of Tab. 6 use injected attack events, and injected times can carry an artifact: the sends of a stealthy carrier arrive at deterministic, evenly

Graph	Timing	KS	probe	vol.	manual	temp.
Enron	injected	0.79	0.76	0.69	**0.88**	0.86
	empirical	0.04	0.63	0.67	0.64	**0.76**
email-Eu-core	injected	0.77	0.71	0.64	**0.85**	0.83
	empirical	0.08	0.54	0.60	0.60	**0.70**
CollegeMsg	injected	0.69	0.70	0.62	**0.82**	0.77
	empirical	0.04	0.56	0.57	0.61	**0.69**

Table 7
Injection-signature control (stealthy regime, 3 seeds; `exp_injection_probe.csv`). KS: two-sample statistic of inter-send gap distributions, attack vs background. Probe: AUC of a purely temporal detector (no graph features). "Empirical" rows re-run the regime with attack times drawn from the background's empirical gap distribution. The receiving-side margin over volume survives the signature's removal only in the learned temporal GNN (bold), not in the hand-crafted features.

Attack frequency	Precision	FP per detected attack
1 : 3 (evaluation)	0.77	0.30
1 : 100	0.09	10
1 : 1000	0.010	101
1 : 10 000	0.001	1,010
$1:10^6$	<0.001	101 000

Table 8
Base-rate arithmetic for the temporal GNN operating point (sensitivity 0.10, FPR1%): at a realistic prevalence of $1:10^4$ each detected attack comes with $\sim$1,010 false alarms, three orders of magnitude above the deployment threshold in FPR terms, which is why the detectors are triage instruments. "FP per detected attack" is $\alpha(1-\pi)/(r\pi)$, where r is sensitivity, α is FPR, and π is prevalence; "$1:n$" is the ratio of attack to legitimate events, i.e. $\pi = 1/(n{+}1)$.

spaced gaps that the background traffic never produces. A two-sample KS test confirms the signature on all three real graphs (statistics 0.46–0.79, $p < 10^{-43}$), and a purely temporal probe—a LightGBM over nothing but inter-send gaps, their regularity, and sender activity, with no graph feature at all—reaches AUC 0.71–0.76 under the stealthy regime: much of the apparent margin rides on *when* the injections happen, not on the receiving-side wave. Tab. 7 therefore repeats the stealthy regime with the attack send times drawn from the background's empirical gap distribution (*empirical* rows): the KS distance collapses to 0.04 ($p \approx 0.5$), the probe to 0.54–0.63, and the hierarchy changes— the learned temporal GNN keeps a margin over volume on all three graphs (0.69–0.76 vs 0.57–0.67), while the hand-crafted features fall back to the volumetric level (0.60–0.64). The honest claim of T1 on real graphs is thus narrower than the raw table suggests: the receiving-side signal survives an attack that paces like the background, but only the learned memory reads it; the hand-crafted context was reading the injection.

Label noise. Flipping an increasing fraction of training labels (Supplementary Tab. S4) shows that the learned GNN degrades fastest (temporal 0.72 → 0.59 at 20% noise, volume only 0.60 → 0.57, hand-crafted 0.66 → 0.61). Under realistic label noise the advantage of the learned model shrinks—another boundary of T2.

Base rate. Sensitivity at a given FPR is independent of prevalence, but precision drops when the attack is rare. From Bayes' theorem $\mathrm{prec}(\pi) = r\pi/(r\pi + \alpha(1-\pi))$ (4): at the operating point of sensitivity r=0.10 at FPR α=1% and a real prevalence of $1:10^4$, each detected attack comes with $\sim$1,010 false alarms (Tab. 8; in the stealthy regime the operating point is weaker still, $r \approx 0.05$, i.e. roughly 2,000 false alarms per detected attack at the same prevalence); in terms of false-positive rate, our operating point (10^{-2}) is about three orders of magnitude ($\sim$2500×) above the threshold of Ho et al. (2019) (< 4/million, i.e. $4 \cdot 10^{-6}$). The consequence is that at a real prevalence our detectors, and any detector with a similar ROC curve, are triage and ranking tools, so the contribution lies in the signal, the mechanism, and the methodology.

$$\mathrm{prec}(\pi) \;=\; \frac{r\,\pi}{r\,\pi + \alpha\,(1-\pi)}, \tag{4}$$

where r is sensitivity, α is FPR, and π is the attack prevalence (base rate).

An operational method meets the standard of Sec. 3.4: its advantage vanishes under the time-shuffle control, transfers across topologies by the predictivity probe, and is measured at low FPR with explicit prevalence arithmetic. By this standard, 98% on a mass benchmark says nothing about lateral phishing. What counts is sensitivity to the targeted tail, with a stated floor (the surgical attack) and an explicit threshold still orders of magnitude away. The next section interprets this gap.

6. Discussion

The cascade invariant addresses the failure mode that sending-side defenses cannot cover, and the rigor analysis marks where the signal stops being operationally useful. We focus here on the mechanism's implications and its limits.

6.1. Interpretation: why the receiving side keeps its margin

A volumetric detector in the spirit of COMPA (Stringhini and Thonnard, 2015) assumes that a compromise shows up as a surge; a path-based detector in the spirit of Hopper (Ho et al., 2021) assumes an unusual route to a rare target. Both read what the attacker controls. Strictly speaking, the path-based baseline is not purely sender-side either: its precursor feature ("the sender was itself reached just before") is the same information as the hand-crafted context's freshness bit, so part of the path signal is a receivership feature already—which the no-precursor ablation of Tab. 5 quantifies. When the attacker deliberately spreads the sends over time, the volumetric detector stays near chance—its event-causal counters do see a burst that builds up within a bucket, but one send per bucket leaves every causal window empty—and the path-based signal degrades as the cascade reaches further. Cascade dynamics instead respond to the recipient side, the wave-like signature among the neighbors, which the sender cannot hide without giving up fan-out altogether—and a single recipient is exactly the surgical regime, which we mark as the blind spot. Pacing the sends weakens this signal (AUC $0.85 \rightarrow 0.59$ from blast to $K=1$), but the volumetric detector never develops a usable signal here in the first place, so the receiving-side signal keeps its margin at every level. This agrees with Ho et al. (Ho et al., 2019) that lateral phishing needs a signal beyond the content of a single message; we differ in locating that signal in the dynamics of the communication graph rather than in headers or relays. The time-shuffle control supports the temporal reading: shuffling removes the temporal order (and, as a side effect, within-bucket co-occurrence), and with it the advantage over the static model.

6.2. Practical implications

For a SOC, the result says what to monitor and where the monitor fails. The receiving-side wave is a property of the organization's own communication graph, computable from headers alone, so it fits an in-house triage pipeline that does not inspect content. The blind spot is equally concrete: a single-recipient transfer request from a compromised account is invisible to every detector studied here, so out-of-band verification of transfers remains the control for that case, and the cascade detector should be paired with it rather than replace it. Two further boundaries shape deployment. The learned model is the most sensitive to noisy training labels, so retraining on analyst-confirmed labels matters more for it than for the volumetric baseline; and at a realistic base rate every detected attack comes with hundreds of false alarms, so the detectors are ranking instruments for analysts, not autonomous blockers. Combining header-only scoring on the graph with content and process controls is the realistic operating mode.

6.3. Limitations

The limitations define the scope of these claims. First, the maliciousness is synthetic. No public corpus links real accounts with labeled compromises and their propagation, so we generate cascades on real structure (Enron, SNAP) or on a synthetic population. We validate the mechanism (a paced cascade separates from legitimate traffic by its receiving-side dynamics), not the detection of real intrusions; the real-graph replications establish the receiving-side signal on real topologies and rhythms, but the injected cascades follow our threat model. Second, the cascade parameters (p_{inf}, depth, fan-out, the stealthy attacker's pacing) are ours; the sweeps in the Supplement show the hierarchy is stable across them, but an attacker who sends to a single recipient (fan-out 1) is outside the signal by construction. Third, the scale is limited ($N=150$ accounts in the case study; real graphs up to about 2k nodes), making the comparison with heavy temporal architectures a sanity check rather than a full benchmark. The case-study graph is also denser than typical corporate mail: every account corresponds with everyone in its firm and only 7% of the contact edges cross firms, so the synthetic cascades meet a compact, well-connected neighborhood. The replications on Enron and the SNAP graphs, whose density and rhythms are real, are what carries the mechanism beyond that setting; the SNAP

replication rests on 3 seeds, so we read it as a regime-level confirmation rather than as a precise effect size. Finally, the operational threshold remains a gap. Even the best operating point (sensitivity@FPR1%) is orders of magnitude from the deployment threshold on real mail (Ho et al. (Ho et al., 2019): < 4 FP/M); we report that gap as the open requirement, not as a solved one.

6.4. Future work directions

The most important next step is validation on a corpus with labeled, real compromises, turning the proof of mechanism into a measurement of detectability. The threat model should then be stressed: an attacker that *learns* the communication graph and tunes both the pacing and the propagation pattern to stay below the receiving-side signal, together with population drift over time (TESSERACT (Pendlebury et al., 2019)). On the modeling side, richer relational context—the organization's hierarchy, competencies, and public footprint as additional layers over the same accounts—is the natural complement for the external impersonation mode, which we study separately. Finally, the receiving-side score is a natural input to a SOC's triage queue, provided an evaluation ties it to a real reduction in analyst load at a fixed alert budget.

7. Conclusions

Modern phishing detection works in aggregate, yet a residual fraction—spear phishing and BEC—remains the most costly and largely evades content filters, and lateral phishing from a compromised account is its hardest case, because every static relation the sender exhibits is genuine. The gap is temporal. By modeling the organization's communication graph with its rhythm, the violation becomes measurable in the dynamics.

We answer the two research questions in turn.

RQ1 (signal, T1): A compromised account consistent with every static relation is detected from the receiving-side wave of its cascade. Unlike volume-based detection, the temporal detector stays above the volumetric one at every stealth level, though both degrade (0.59 vs 0.48 at $K{=}1$; Section 5.4); the advantage vanishes under a time-shuffle control; the regime contrast—volume wins the naive blast, the receiving-side signal wins the stealthy regime—replicates on Enron and two SNAP graphs, and the injected-timing control bounds the stealth margin to the learned temporal model (Section 5.7).

RQ2 (limits, T2): The signal needs fan-out: a single-recipient attack is invisible to every detector (Section 5.5). The advantage degrades gracefully with scale and shrinks under label noise, and at a realistic base rate the detectors act as triage instruments, with effectiveness defined operationally through low-FPR sensitivity under ablation controls and explicit prevalence arithmetic (Section 5.8).

As its contributions, the paper offers a formalization of lateral phishing as a temporal violation of the organizational communication graph, with a cascade threat model that includes the stealthy attacker; the receiving-side cascade signal as a detection invariant, with a hierarchy of detectors, a time-shuffle control, and validation on three real communication graphs; and a leak-aware evaluation methodology with an operational definition of detection effectiveness for the targeted tail.

Among the limitations, the maliciousness is synthetic, so we validate the mechanism rather than the detection of real intrusions; the cascade parameters are ours; and both scale and the operational threshold remain open. Section 6 gives the details.

The results turn lateral phishing into a quantity that can be tracked rather than merely flagged. Under the default cascade the receiving-side signal reaches AUC 0.87; under a stealthy attacker it degrades to 0.59–0.76 but stays above volume, which remains near chance in the stealthy regime (0.48–0.64) and only recovers under the naive blast; and the same regime contrast appears on Enron, `email-Eu-core`, and `CollegeMsg`—where the injected-timing control narrows the honest claim: with attack times drawn from the background, only the learned temporal model keeps the receiving-side margin over volume. The defense this points toward asks not whether a message looks malicious, but whether the organization's communication graph is behaving the way it usually does. Turning that question into an operational filter in the SOC pipeline, at the precision real deployment demands, remains an open problem.

CRediT authorship contribution statement

Kamil Warpechowski: Conceptualization, Methodology, Software, Investigation, Validation, Writing – original draft. **Bogdan Księżopolski:** Supervision, Conceptualization, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The reproducibility package, including the synthetic population generator, all experiment and analysis code, the generated population of 150 personas, and the computed results from which the tables and figures are produced, is openly archived on Zenodo (DOI: to be inserted upon deposit). The real communication graphs used for validation are public: the Enron email corpus (Klimt and Yang, 2004) and the SNAP datasets `email-Eu-core` and `CollegeMsg` (Leskovec and Krevl, 2014).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to translate the manuscript from Polish into English and to copy-edit the prose. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Abdelnabi, S., Krombholz, K., Fritz, M., 2020. VisualPhishNet: Zero-day phishing website detection by visual similarity, in: ACM CCS, pp. 1681–1698. doi:10.1145/3372297.3417233.

Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L., Rieck, K., 2022. Dos and don'ts of machine learning in computer security, in: 31st USENIX Security Symposium. doi:10.48550/arXiv.2010.09470.

Atlam, H.F., Oluwatimilehin, O., 2023. Business email compromise phishing detection based on machine learning: A systematic literature review. Electronics 12. doi:10.3390/electronics12010042.

Behrouz, A., Seltzer, M., 2022. Anomaly detection in multiplex dynamic networks: from blockchain security to brain disease prediction. arXiv:2211.08378.

Bethany, M., Galiopoulos, A., Bethany, E., Karkevandi, M.B., Vishwamitra, N., Najafirad, P., 2024. Evaluating large language models' capability to launch fully automated spear phishing campaigns: Validated on human subjects. arXiv:2412.00586.

Bian, T., Xiao, X., Xu, T., Zhao, P., Huang, W., Rong, Y., 2020. Rumor detection on social media with bi-directional graph convolutional networks, in: Proceedings of the AAAI Conference on Artificial Intelligence. doi:10.1609/aaai.v34i01.5393.

Birthriya, S.K., Ahlawat, P., Jain, A.K., 2025. Detection and prevention of spear phishing attacks: A comprehensive survey. Computers & Security 151. doi:10.1016/j.cose.2025.104317.

Bountakas, P., Xenakis, C., 2022. HELPHED: Hybrid ensemble learning PHishing email detection. Journal of Network and Computer Applications 210. doi:10.1016/j.jnca.2022.103545.

Butavicius, M., Taib, R., Han, S.J., 2022. Why people keep falling for phishing scams: The effects of time pressure and deception cues on the detection of phishing emails. Computers & Security 123. doi:10.1016/j.cose.2022.102937.

Cheng, J., Adamic, L., Dow, P.A., Kleinberg, J.M., Leskovec, J., 2014. Can cascades be predicted?, in: Proceedings of the 23rd International Conference on World Wide Web. doi:10.1145/2566486.2567997.

Cidon, A., Gavish, L., Bleier, I., Korshun, N., Schweighauser, M., Tsitkin, A., 2019. High precision detection of business email compromise, in: 28th USENIX Security Symposium.

Ding, K., Li, J., Bhanushali, R., Liu, H., 2019. Deep anomaly detection on attributed networks, in: SIAM SDM. doi:10.1137/1.9781611975673.67.

Dou, Y., Liu, Z., Sun, L., Deng, Y., Peng, H., Yu, P.S., 2020. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters, in: ACM CIKM. doi:10.1145/3340531.3411903.

Duman, S., Kalkan-Cakmakci, K., Egele, M., Robertson, W., Kirda, E., 2016. EmailProfiler: Spearphishing filtering with header and stylometric features of emails, in: IEEE COMPSAC. doi:10.1109/COMPSAC.2016.105.

Egele, M., Stringhini, G., Kruegel, C., Vigna, G., 2013. COMPA: Detecting compromised accounts on social networks, in: Network and Distributed System Security Symposium (NDSS).

Federal Bureau of Investigation, 2024. Internet crime report 2023. URL: https://www.ic3.gov/Media/PDF/AnnualReport/2023_IC3Report.pdf.

Gao, Y., Wang, X., He, X., Liu, Z., Feng, H., Zhang, Y., 2023. Alleviating structural distribution shift in graph anomaly detection, in: ACM WSDM. doi:10.1145/3539597.3570377.

Gascon, H., Ullrich, S., Stritter, B., Rieck, K., 2018. Reading between the lines: Content-agnostic detection of spear-phishing emails, in: RAID. doi:10.1007/978-3-030-00470-5_4.

Guo, B., Wang, S., Yue, T., Sun, Y., Hu, X., 2025. Efficient phishing URL detection using graph-based machine learning and loopy belief propagation. arXiv:2501.06912.

Hamilton, W.L., Ying, R., Leskovec, J., 2017. Inductive representation learning on large graphs, in: NeurIPS. arXiv:1706.02216.

Hao, Y., Cao, X., Fang, Y., Xie, X., Wang, S., 2020. Inductive link prediction for nodes having only attribute information, in: IJCAI. arXiv:2007.08053.

Ho, G., Cidon, A., Gavish, L., Schweighauser, M., Paxson, V., Savage, S., Voelker, G.M., Wagner, D., 2019. Detecting and characterizing lateral phishing at scale, in: 28th USENIX Security Symposium. arXiv:1910.00790.

Ho, G., Dhiman, M., Akhawe, D., Paxson, V., Savage, S., Voelker, G.M., Wagner, D., 2021. Hopper: Modeling and detecting lateral movement, in: 30th USENIX Security Symposium. arXiv:2105.13442.

Ho, G., Sharma, A., Javed, M., Paxson, V., Wagner, D., 2017. Detecting credential spearphishing in enterprise settings, in: 26th USENIX Security Symposium.

Klimt, B., Yang, Y., 2004. The Enron corpus: A new dataset for email classification research, in: ECML. doi:10.1007/978-3-540-30115-8_22.

Kumar, S., Zhang, X., Leskovec, J., 2019. Predicting dynamic embedding trajectory in temporal interaction networks, in: ACM KDD. doi:10.1145/3292500.3330895.

Leskovec, J., Krause, A., Guestrin, C., Faloutsos, C., VanBriesen, J., Glance, N., 2007. Cost-effective outbreak detection in networks, in: Proceedings of the 13th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. doi:10.1145/1281192.1281239.

Leskovec, J., Krevl, A., 2014. SNAP Datasets: Stanford large network dataset collection. http://snap.stanford.edu/data.

Li, S., Gou, G., Liu, C., Hou, C., Li, Z., Xiong, G., 2022a. TTAGN: Temporal transaction aggregation graph network for Ethereum phishing scams detection, in: The Web Conference (WWW). doi:10.1145/3485447.3512226.

Li, S., Xu, G., Xie, R., Zhou, L., Xuan, Q., 2022b. Phishing fraud detection on Ethereum using graph neural network. arXiv:2204.08194.

Li, X., Qi, J., Zhao, Z., Zheng, Y., Cao, L., Dong, J., Yu, G., 2025. UMGAD: Unsupervised multiplex graph anomaly detection, in: IEEE ICDE. arXiv:2411.12556.

Lin, Y., Liu, R., Divakaran, D.M., Dong, J.Y., Ng, G., Jiang, H., Chen, C., Zhang, Y., Lau, Y., Dong, J.S., 2021. Phishpedia: A hybrid deep learning based approach to visually identify phishing webpages, in: 30th USENIX Security Symposium.

Liu, Y., Ao, X., Qin, Z., Chi, J., Feng, J., Yang, H., He, Q., 2021a. Pick and choose: A GNN-based imbalanced learning approach for fraud detection, in: The Web Conference (WWW). doi:10.1145/3442381.3449989.

Liu, Y., Li, Z., Pan, S., Gong, C., Zhou, C., Karypis, G., 2021b. Anomaly detection on attributed networks via contrastive self-supervised learning. IEEE Transactions on Neural Networks and Learning Systems doi:10.1109/TNNLS.2021.3068344.

Liu, Z., Dou, Y., Yu, P.S., Deng, Y., Peng, H., 2020. Alleviating the inconsistency problem of applying graph neural network to fraud detection, in: ACM SIGIR. doi:10.1145/3397271.3401253.

Luo, S., Young, E., Ho, G., Schweighauser, M., Katz-Bassett, E., Cidon, A., 2024. Characterizing the networks sending enterprise phishing emails. arXiv:2412.12403.

Ma, X., Wu, J., Xue, S., Yang, J., Zhou, C., Sheng, Q.Z., Xiong, H., Akoglu, L., 2021. A comprehensive survey on graph anomaly detection with deep learning. IEEE Transactions on Knowledge and Data Engineering 35. doi:10.1109/TKDE.2021.3118815.

Nayebi Kerdabadi, M., Byron, A., Sun, Z., Iranitalab, A., 2025. Spatio-temporal directed graph learning for account takeover fraud detection. arXiv:2509.20339.

Nayeri, Z.M., Rezvani, M., 2026. Alert prediction in computer networks using transformer-based temporal graph neural networks: Identifying the next victim. Journal of Network and Computer Applications 249. doi:10.1016/j.jnca.2026.104455.

Panzarasa, P., Opsahl, T., Carley, K.M., 2009. Patterns and dynamics of users' behavior and interaction: Network analysis of an online community. Journal of the American Society for Information Science and Technology 60, 911–932. doi:10.1002/asi.21015.

Paranjape, A., Benson, A.R., Leskovec, J., 2017. Motifs in temporal networks, in: Proceedings of the Tenth ACM International Conference on Web Search and Data Mining (WSDM). doi:10.1145/3018661.3018731.

Pareja, A., Domeniconi, G., Chen, J., Ma, T., Suzumura, T., Kanezashi, H., Kaler, T., Schardl, T.B., Leiserson, C.E., 2020. EvolveGCN: Evolving graph convolutional networks for dynamic graphs, in: AAAI. arXiv:1902.10191.

Park, C., Kim, D., Han, J., Yu, H., 2020. Unsupervised attributed multiplex network embedding, in: AAAI. arXiv:1911.06750.

Pendlebury, F., Pierazzi, F., Jordaney, R., Kinder, J., Cavallaro, L., 2019. TESSERACT: Eliminating experimental bias in malware classification across space and time, in: 28th USENIX Security Symposium. arXiv:1807.07838.

Pinto, P.C., Thiran, P., Vetterli, M., 2012. Locating the source of diffusion in large-scale networks. Physical Review Letters 109, 068702. doi:10.1103/PhysRevLett.109.068702.

Priebe, C.E., Conroy, J.M., Marchette, D.J., Park, Y., 2005. Scan statistics on Enron graphs. Computational and Mathematical Organization Theory 11, 229–247. doi:10.1007/s10588-005-4094-7.

Rayana, S., Akoglu, L., 2015. Collective opinion spam detection: Bridging review networks and metadata, in: ACM KDD. doi:10.1145/2783258.2783370.

Rossi, E., Chamberlain, B., Frasca, F., Eynard, D., Monti, F., Bronstein, M., 2020. Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637.

Safran, M., Musleh, S., 2025. PhishingGNN: Phishing email detection using graph attention networks and transformer-based feature extraction. IEEE Access 13. doi:10.1109/ACCESS.2025.3592135.

Sankar, A., Wu, Y., Gou, L., Zhang, W., Yang, H., 2020. DySAT: Deep neural representation learning on dynamic graphs via self-attention networks, in: ACM WSDM. doi:10.1145/3336191.3371845.

Shah, D., Zaman, T., 2011. Rumors in a network: Who's the culprit? IEEE Transactions on Information Theory 57, 5163–5181. doi:10.1109/TIT.2011.2158885.

Shi, F., Cao, Y., Shang, Y., Zhou, Y., Zhou, C., Wu, J., 2022. H2-FDetector: A GNN-based fraud detector with homophilic and heterophilic connections, in: The Web Conference (WWW). doi:10.1145/3485447.3512195.

Stringhini, G., Thonnard, O., 2015. That ain't you: Blocking spearphishing through behavioral modelling, in: DIMVA. doi:10.1007/978-3-319-20550-2_5.

Sturman, D., Auton, J.C., Morrison, B.W., 2025. Security awareness, decision style, knowledge, and phishing email detection: Moderated mediation analyses. Computers & Security 148. doi:10.1016/j.cose.2024.104129.

Tan, C.L., Chiew, K.L., Yong, K.S., Sze, S.N., Abdullah, J., Sebastian, Y., 2020. A graph-theoretic approach for the detection of phishing webpages. Computers & Security 95. doi:10.1016/j.cose.2020.101793.

Tang, J., Li, J., Gao, Z., Li, J., 2022. Rethinking graph neural networks for anomaly detection, in: ICML. arXiv:2205.15508.

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y., 2018. Graph attention networks, in: ICLR. arXiv:1710.10903.

Vosoughi, S., Roy, D., Aral, S., 2018. The spread of true and false news online. Science 359, 1146–1151. doi:10.1126/science.aap9559.

Wang, J., Chen, P., Xu, X., Wu, J., Shen, M., Xuan, Q., Yang, X., 2022. TSGN: Transaction subgraph networks assisting phishing detection in Ethereum. arXiv:2208.12938.

Weber, M., Domeniconi, G., Chen, J., Weidele, D.K.I., Bellei, C., Robinson, T., Leiserson, C.E., 2019. Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics. arXiv:1908.02591.

Weng, Z., Zhang, W., Zhu, T., Dou, Z., Sun, H., Ye, Z., Tian, Y., 2024. RT-APT: A real-time APT anomaly detection method for large-scale provenance graph. Journal of Network and Computer Applications 233. doi:10.1016/j.jnca.2024.104036.

Wu, J., Yuan, Q., Lin, D., You, W., Chen, W., Chen, C., Zheng, Z., 2020. Who are the phishers? phishing scam detection on Ethereum via network embedding. IEEE Transactions on Systems, Man, and Cybernetics: Systems doi:10.1109/TSMC.2020.3016821.

Xu, D., Ruan, C., Korpeoglu, E., Kumar, S., Achan, K., 2020. Inductive representation learning on temporal graphs, in: ICLR. arXiv:2002.07962.

You, J., Du, T., Leskovec, J., 2022. ROLAND: Graph learning framework for dynamic graphs, in: ACM KDD. doi:10.1145/3534678.3539300.

Yu, S., Xia, W., Liu, B., Wu, J., 2023. Streaming phishing scam detection method on Ethereum. arXiv:2306.16624.

Zhang, D., Chen, J., 2021. Blockchain phishing scam detection via multi-channel graph classification. arXiv:2108.08456.

Zhang, H., Qiu, L., Yi, L., Song, Y., 2018. Scalable multiplex network embedding, in: IJCAI. doi:10.24963/ijcai.2018/428.

Zhang, M., Chen, Y., 2018. Link prediction based on graph neural networks, in: NeurIPS. doi:10.48550/arXiv.1802.09691.

Zhao, T., Zhang, X., Wang, S., 2021. GraphSMOTE: Imbalanced node classification on graphs with graph neural networks, in: ACM WSDM. doi:10.1145/3437963.3441720.